Rahul Singh

AI Engineer — Data Scientist — Generative AI — Agentic AI — LLM Applications

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Profile

Data Scientist with 3+ years of experience in the healthcare and automotive sectors, specializing in statistical modeling, machine learning, deep learning, and Generative AI. Proven ability to design and deploy end-to-end predictive pipelines and LLM applications utilizing Docker and AWS to solve complex business challenges.

Technical Skills

- **Programming Data Analysis:** Python , Pandas , NumPy , PySpark, SQL.
- **Generative AI NLP:** NLP, Generative AI , OpenAI (ChatGPT), Google Gemini, LangChain , Prompt Engineering , Vector Search, RAG , MCP, Agentic-ai.
- **AI Engineering Agentic AI:** LangGraph, LlamaIndex, AI Agents, Multi-Agent Systems, Workflow Orchestration, Tool Calling, Agentic AI, MCP.
- **Machine Learning Deep Learning:** Recommendation Systems, ML/DL model development , ANN , CNN , RNN , LSTM , Transformers.
- **Statistical Analysis EDA:** Univariate, bivariate, multivariate analysis , feature engineering
- **Databases:**MySQL , PostgreSQL , Oracle , MongoDB , Amazon Athena
- **Deployment Cloud:** Docker , GitHub , AWS (S3, ECS,ECR CloudWatch)

Professional Experience

Data Scientist *(NITS)* **Gurgaon, Haryana, India** *02/2025 - Present*

- Designed and developed AI Agents using LangGraph for multi-step reasoning and workflow orchestration.
- Built enterprise-grade RAG applications using LlamaIndex and vector databases to enable context-aware information retrieval and question answering.
- Developed Agentic AI workflows integrating LLMs, external tools, APIs, and databases to automate complex business processes and decision-making tasks.
- Utilized Python to perform advanced statistical analysis, EDA, and feature engineering on structured datasets of 10M+ records, ensuring high-quality model inputs.
- Architected and deployed Generative AI and NLP workflows, integrating OpenAI (ChatGPT) and Google Gemini models to automate internal data processing.
- Applied rigorous Prompt Engineering and LLM fine-tuning techniques, resulting in a 20 percent improvement in answer relevance and clarity.
- Leveraged Amazon Athena and Oracle for large-scale data querying to support scalable analytics and predictive model development.

Data Scientist *(Althea.ai)* **Noida, UP, India** *05/2022 - 1/2024*

- Led data preprocessing, feature engineering, and exploratory data analysis (EDA) using Python (Pandas, NumPy), translating complex datasets into clear statistical visualizations using Matplotlib and Seaborn.
- Developed and trained advanced machine learning and deep learning models utilizing TensorFlow, translating statistical insights into high-impact predictive solutions that improved core business metrics by 7 percent
- Optimized predictive models via rigorous hyperparameter tuning (Grid Search) and Random Forest ensemble techniques, maximizing F1-score and recall while significantly enhancing production stability.
- Containerized machine learning models and data processing pipelines utilizing Docker, establishing scalable, cross-environment consistency that accelerated production deployments.
- Engineered robust cloud-based infrastructure on AWS, leveraging Amazon S3 for secure data management, orchestrating scalable deployments via ECS, and establishing system monitoring with CloudWatch.

Projects

AI VIsualizer LLM Powered Data Visualization Agent *(NITS)* **Gurgaon, Haryana, India**

- Developed an end to end AI-Visualization agent that convert the nlp request and structured data into validated, Presentation-ready Chart.
- Design a multi-Stage LLM workflow with separate agents for visualization planning and Python Chart code generation .
- Build a reusable visualization pipeline convering request validation, default configuration, chart planning, code generation, secure execution and structured response handling
- Developed an MCP server exposing visualization functionality through tools and resources for integration with external AI-Agents.
- Integrated The MCP server with a REST API framework and Server-Sent-Event transport for remote communication and service monitoring.

Chronic Kidney Disease *(Althea.ai)* **Noida, UttarPradesh, India**

Developed a healthcare data pipeline using Python and MySQL to extract and preprocess clinical records for predictive modeling.

Trained advanced classification models (XGBoost, CatBoost) and applied NLP techniques to process unstructured medical notes for better feature extraction.

Trained and evaluated advanced classification models (XGBoost, CatBoost, Random Forest, SVM) to accurately predict disease onset.

Integrated systematic logging and exception handling throughout the model training framework to ensure pipeline reliability and data integrity.

Deployed the final predictive model as a scalable, Dockerized Flask web application on Amazon Web Services (AWS).

Education

Computer Science Engineering *Abdul Kalam Technical University* **U.P, India** *2017-2021*

Relevant Courses: Computer Science Engineering